

The Cubic Surface Discriminant in Clebsch–Salmon Invariants

Let $S \subset \mathbf{P}^3$ be a cubic surface over a field of characteristic zero, and let

$$(A, B, C, D, E) = (I_8, I_{16}, I_{24}, I_{32}, I_{40})$$

denote the Clebsch–Salmon invariants in the standard normalization. The Clebsch-normalized discriminant is the following weighted homogeneous polynomial of degree 32 in the coefficients of the cubic surface.

Proposition. *With the above normalization,*

$$\Delta_{\text{Cl}} = (A^2 - 64B)^2 - 2^{11}(8D + AC).$$

Equivalently,

$$\Delta_{\text{Cl}} = A^4 - 128A^2B + 4096B^2 - 2048AC - 16384D.$$

In particular, the invariant E does not enter the formula.

For comparison with the pentahedral form, write a general cubic surface as

$$S_{(a_0, \dots, a_4)} : \quad \sum_{i=0}^4 a_i X_i^3 = 0, \quad \sum_{i=0}^4 X_i = 0,$$

and let σ_j be the elementary symmetric polynomial of degree j in $a_0, \dots, a_4$. In this convention the fundamental invariants are

$$A = \sigma_4^2 - 4\sigma_3\sigma_5, \quad B = \sigma_1\sigma_5^3, \quad C = \sigma_4\sigma_5^4, \quad D = \sigma_2\sigma_5^6, \quad E = \sigma_5^8.$$

Substitution gives exactly the discriminant formula above; this is Lemma 2.6 of Elsenhans–Jahnel [1]. They also note that the formula is classical, due to Salmon, but that the version printed by Salmon contained a misprint later corrected by Edge [2].

Thus, in this normalization, S is singular precisely when

$$\Delta_{\text{Cl}} = 0.$$

If instead one uses the resultant-normalized discriminant of the quaternary cubic form, the expression above must be multiplied by the fixed nonzero scalar dictated by that normalization.

References

- [1] A.-S. Elsenhans and J. Jahnel, *The discriminant of a cubic surface*, Geom. Dedicata **159** (2012), 29–40. doi:10.1007/s10711-011-9643-7.
- [2] W. L. Edge, *The discriminant of a cubic surface*, Proc. Roy. Irish Acad. Sect. A **80** (1980), 75–78.